Choose the best answer for the following questions

- 1. Define the process as________.

A. a program

B. a job

C. a program code

D. a program in execution

- 2. What is the possible transition from a process state to another?

A. new->terminated

B. run->waiting

C. waiting ->running

D. run-> new

- 3. Consider a computer with only one CPU inside, actually how many jobs are executed by the CPU?

A. 1

B. 2

C. 3

D. N

- 4. Given a system with n process, how many possible ways can those process be scheduled?

A. n-2

B. n-1

C. n

D. n+1

- 5 If a process is waiting to be assigned to a processor, it must be in the state of ______.

  A ready

  B waiting

  C running

  D terminated

- 6 In UNIX, a new process is created by the ______ system call.

  A exit()

  B wait

  C exec()

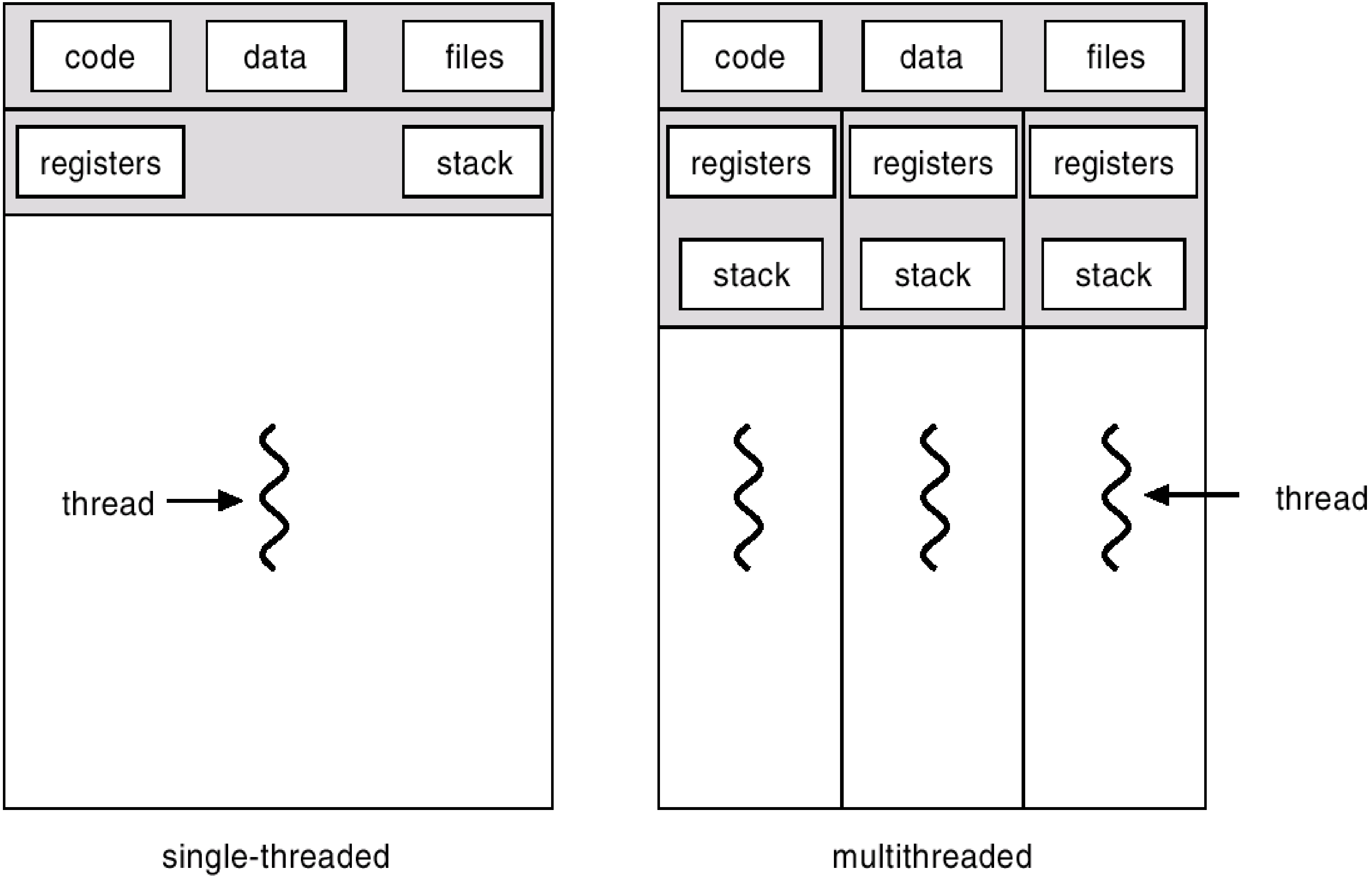
  D fork()

- Which is the method used to pass parameters between a running program and the operating system ?

Chapter 4: Threads(线程)

- Overview
- Multithreading Models(多线程模型)
- Threading Issues
- PThreads
- Windows Threads
- Linux Threads
- Java Threads

Single and Multithreaded Processes



Benefits

- Responsiveness
- Resource Sharing(资源共享)
- Economy
- Utilization of MP Architectures

User Threads(用户线程)

- Thread management done by user-level threads library
- Three primary thread libraries:
  - POSIX Pthreads
  - Win32 threads
  - Java threads

Kernel Threads(内核线程)

- Supported by the Kernel
- Examples

  - Windows
  - Solaris
  - Tru64 UNIX
  - BeOS
  - Linux

Multithreading Models(多线程模型)

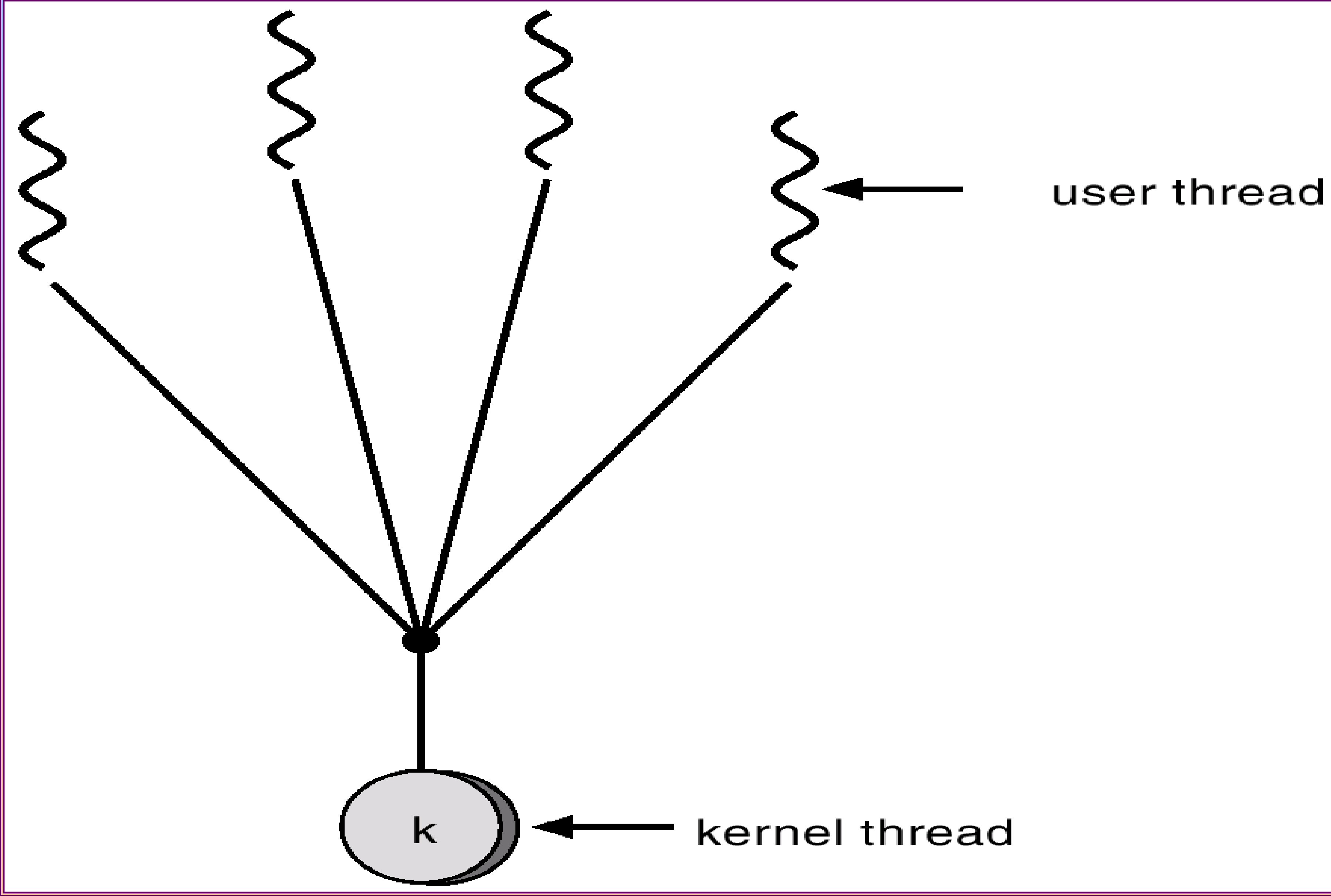
- Many-to-One
- One-to-One
- Many-to-Many

(user thread - to - kernel thread)

Many-to-One(多对一线程模型)

- Many user-level threads mapped to single kernel thread.
- Used on systems that do not support kernel threads.

Many-to-One(多对一线程模型)



One-to-One(一对一线程模型)

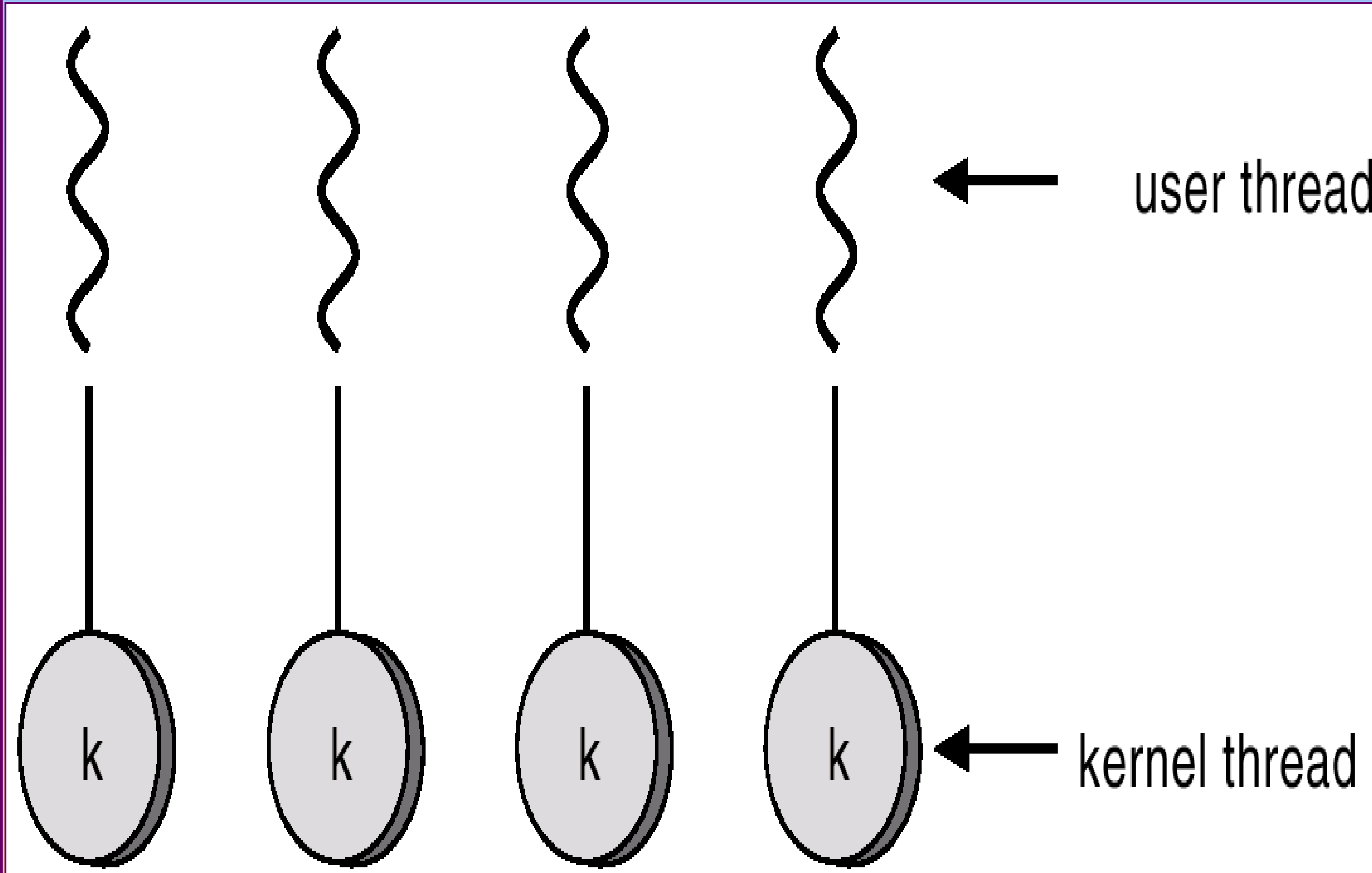
- Each user-level thread maps to kernel thread.

- Examples

  \- Windows

  \- OS/2

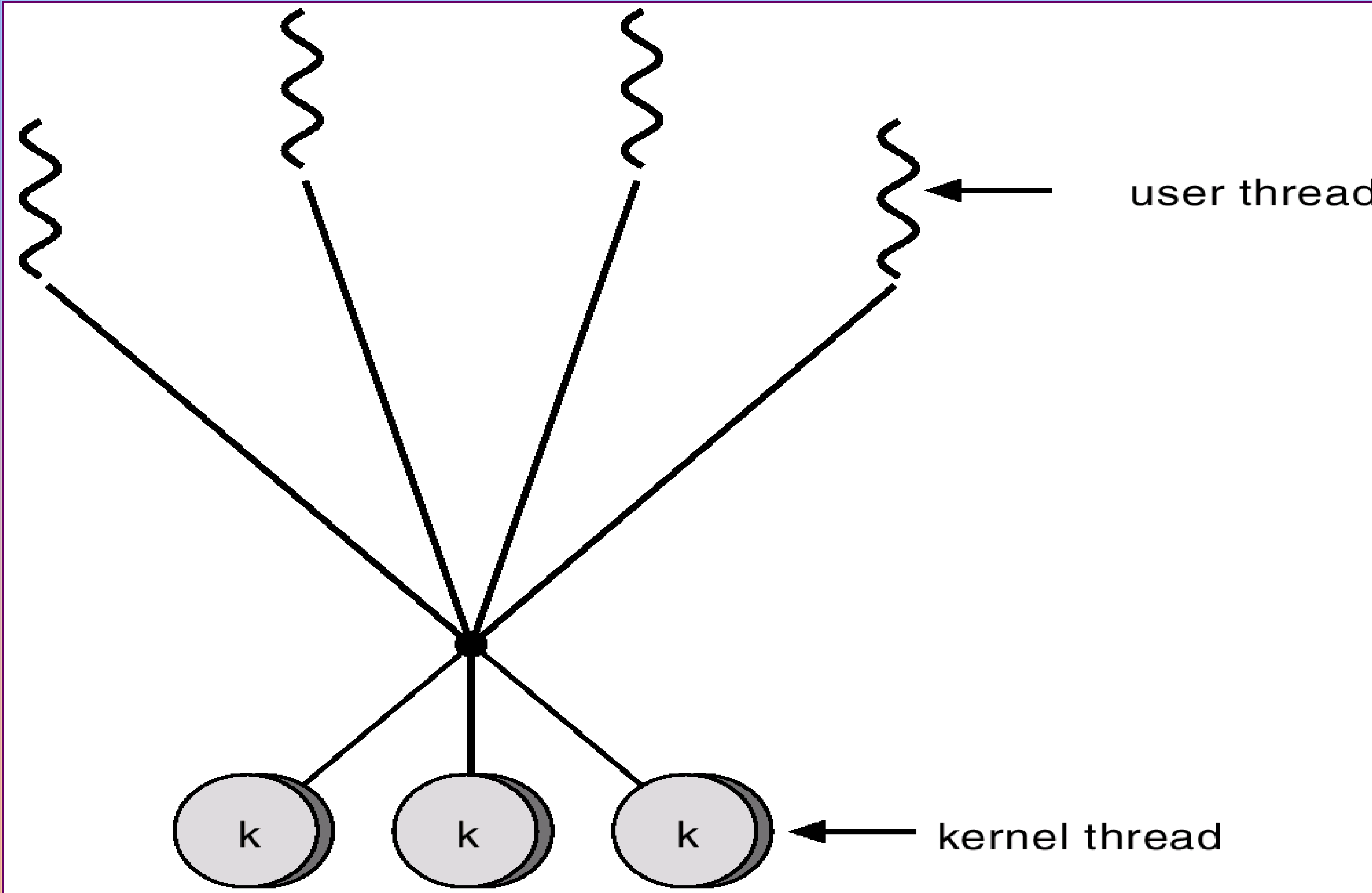
One-to-One(一对一线程模型)



Many-to-Many Model(多读多线程模型)

- Allows many user level threads to be mapped to many kernel threads.
- Allows the operating system to create a sufficient number of kernel threads.
- Solaris 2
- Windows NT/2000 with the *ThreadFiber* package

Many-to-Many Model(多读多线程模型)



Threading Issues

- Semantics of fork() and exec() system calls.
- Thread cancellation.
- Signal handling
- Thread pools
- Thread specific data

Pthreads

- A POSIX standard (IEEE 1003.1c) API for thread creation and synchronization
- API specifies behavior of the thread library, implementation is up to development of the library
- Common in UNIX operating systems (Solaris, Linux, Mac OS X)

Windows Threads

- Implements the one-to-one mapping
- Each thread contains
  - A thread id
  - Register set
  - Separate user and kernel stacks
  - Private data storage area
- The register set, stacks, and private storage area are known as the **context(上下文)** of the threads

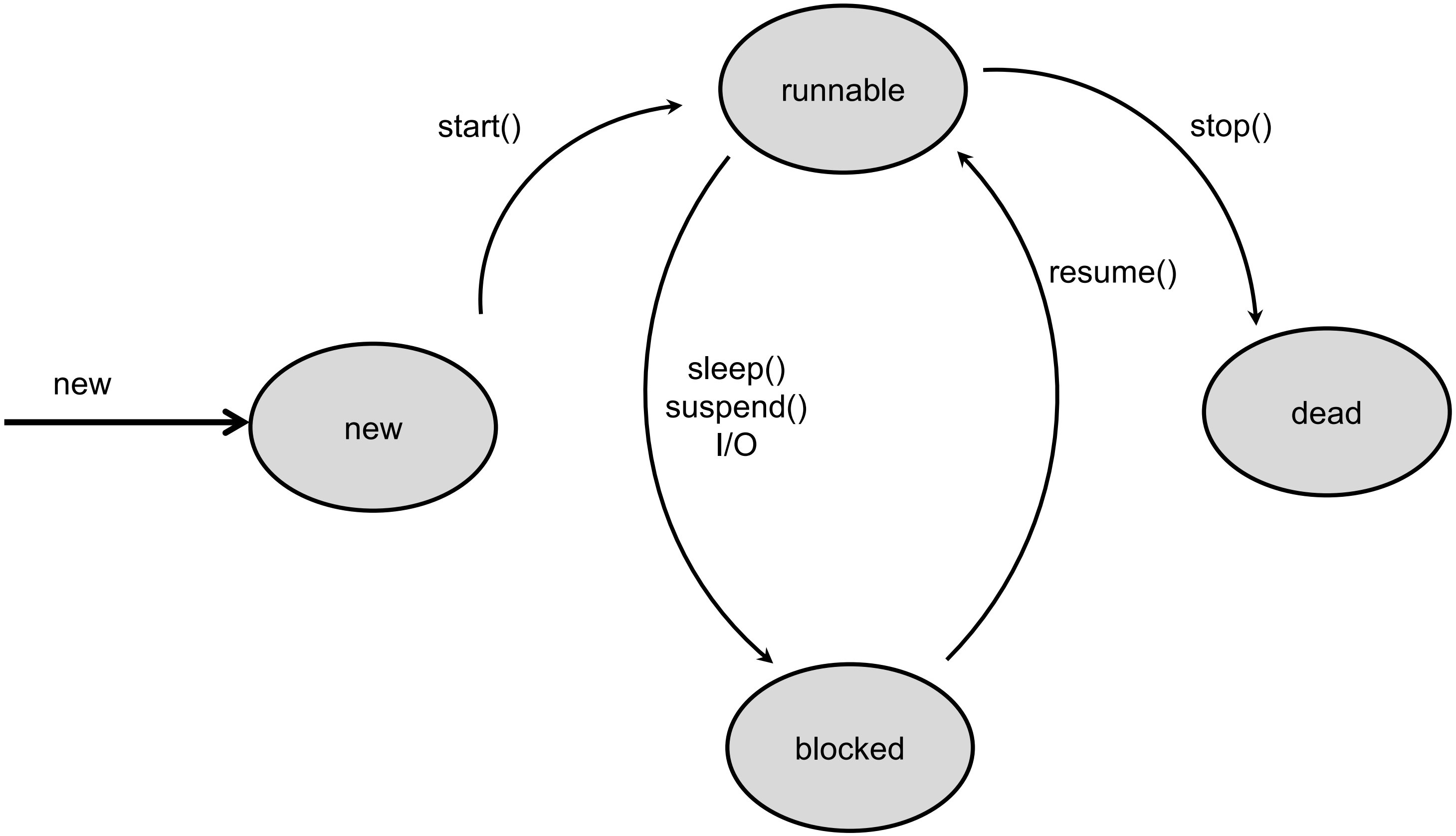
Linux Threads

- Linux refers to them as *tasks* rather than *threads*.
- Thread creation is done through clone() system call.
- Clone() allows a child task to share the address space of the parent task (process)

Java Threads

- Java threads may be created by:
  - Extending Thread class
  - Implementing the Runnable interface
- Java threads are managed by the JVM.

Java Thread States



homework

- 4.7
- 4.11